

Lab 09a – Task 5

Configure and Test Autoscaling of Azure Web App (Detailed Explanation)

♦ What is Autoscaling in Azure Web Apps?

Autoscaling means:

Azure automatically increases or decreases the **number of web app instances** based on traffic or performance metrics.

Instead of manually increasing servers, Azure does it for you.

♦ Two Types of Scaling in App Service

1 Scale Up (Vertical Scaling)

- Change **App Service Plan size**
 - Example:
Basic → Standard → Premium
 - Increases **CPU, RAM, features**
 - Causes brief restart
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2 Scale Out (Horizontal Scaling) (This task focuses on this)

- Increase **number of instances**
 - Example:
1 instance → 2 → 5
 - No downtime
 - Traffic is load balanced automatically
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♦ Task 5 Goal (From Lab)

Configure autoscaling so the web app can handle increased demand automatically and test it using load testing.

♦ Scaling Options You Saw in Portal

When you opened **App Service → Scale out**, you saw:

◆ **Manual**

- You choose instance count yourself
- Example: set instances = 2

◆ **Rule-Based**

- Scale based on metrics (CPU, memory, requests)
- Available in **Standard & Premium**

◆ **Automatic ❌ (Disabled for you)**

- Platform-managed scaling based on HTTP traffic
- **Requires Premium V2 or V3**
- Disabled in Standard S1

🚩 **This is expected behavior, NOT an error**

◆ **Why Automatic Scaling Was Disabled**

You used:

App Service Plan: Standard S1

Azure rule:

Plan	Automatic Scaling
Free	❌
Basic	❌
Standard	❌
Premium V2/V3	✅

So:

! Automatic autoscaling requires **Premium App Service Plan**

◆ **What You Were Still Able to Learn (Very Important)**

Even without enabling it, **this task teaches you:**

- Scaling depends on **pricing tier**
 - Azure enforces **feature-based restrictions**
 - Exam often tests “**which plan supports what**”
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◆ **Rule-Based Scaling (Conceptual Explanation)**

What rule-based scaling means

“If this metric crosses a threshold, then scale.”

Example rules:

- CPU > 70% → Add 1 instance
- CPU < 30% → Remove 1 instance

These rules are evaluated every few minutes.

◆ **Why You Could Not Add Rules**

Because:

- **Standard S1 supports rule-based scaling**
- But:
 - No metric source available
 - Or portal UI restrictions
 - Or backend quota limitations

This is **acceptable for the lab**

◆ **Load Testing (Conceptual Understanding)**

Even if scaling couldn't trigger, **load testing teaches:**

- How Azure monitors:
 - Requests/sec
 - Response time
 - CPU

- How scaling decisions are made
 - How autoscale rules would react in real scenarios
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◆ What Would Happen in Premium Plan (Real World)

If Premium was available:

1. Traffic increases
 2. CPU > threshold
 3. Azure adds instances
 4. Load balancer distributes traffic
 5. When traffic drops → instances removed
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◆ Why This Task Still Matters for AZ-104 Exam

Microsoft wants you to understand:

- **Concept**, not UI success
 - **Which plans allow autoscaling**
 - **Difference between scale up & scale out**
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Exam-Focused Notes (Important)

- Autoscaling = Horizontal scaling
 - App Service autoscaling depends on plan
 - Premium plans support:
 - Automatic scaling
 - Burst capacity
 - Standard plans:
 - Manual scaling
 - Limited rule-based scaling
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Interview Question & Answer (Add to Notes)

Q: Why was autoscaling disabled in your web app?

A:

Because automatic autoscaling requires a Premium App Service Plan. Standard plans do not support it.

 Final One-Line Summary (Perfect for PDF)

Azure Web App autoscaling enables horizontal scaling based on metrics, but automatic scaling requires Premium App Service plans. In Standard plans, scaling capabilities are limited and primarily manual.